ECE 340 # Homework Assignment #5 **Fall 2026**

DUE: FRIDAY, OCTOBER 2, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN ON TIME BEFORE 5:00PM ON THE DUE DATE.

1. STEADY STATE CARRIER GENERATION; QUASI-FERMI LEVEL

A Si sample is doped with $N_D = 2 * 10^{16}$ cm^{-3} and optically excited uniformly with $g_{op} = 10^{19}$ EHP/cm^3·s. Assume T = 300 K and use thermal generation rate $g(T) = 4.5 \times 10^{10} cm^{-3}$/s and diffusion coefficients $D_n = 30$ cm^2/s and $D_p = 10$ cm^2/s.

(A). Find the thermal equilibrium electron and hole concentrations n_0 and p_0.

(B). State the low-level injection and steady-state band-to-band recombination assumptions. Using these assumptions, simplify the generation-recombination balance equation $g(T) + g_{op} = \alpha_r np$ into an expression for the excess carrier concentration δp.

(C). Solve for δp.

(D). Determine the steady-state electron hole concentrations n and p. Check that the low-level injection assumption from (B) is consistent with your answer.

(E). Calculate the separation of the electron and hole quasi-Fermi levels $(F_n - F_p)$.

(F). Calculate the change in conductivity of the Si upon illumination.

2. DIFFUSION AND DRIFT OF CARRIERS

Consider the **thermal equilibrium** band diagram of Figure 2.1 for a portion of a semiconductor sample with a built-in electric field $\boldsymbol{E}$.

(A). Sketch the Fermi level as a function of position *through* the indicated point, E_F, across the width of the band diagram of Figure 2.1.

(B). Based on Figure 2.1, sketch and label both logarithmic electron and hole concentrations as a function of position across the full width of the sample. Label the slopes for each and the location of $\log(n_i)$. *Note:* the carrier concentration scale should be $\log(n), \log(p)$ on the same graph.

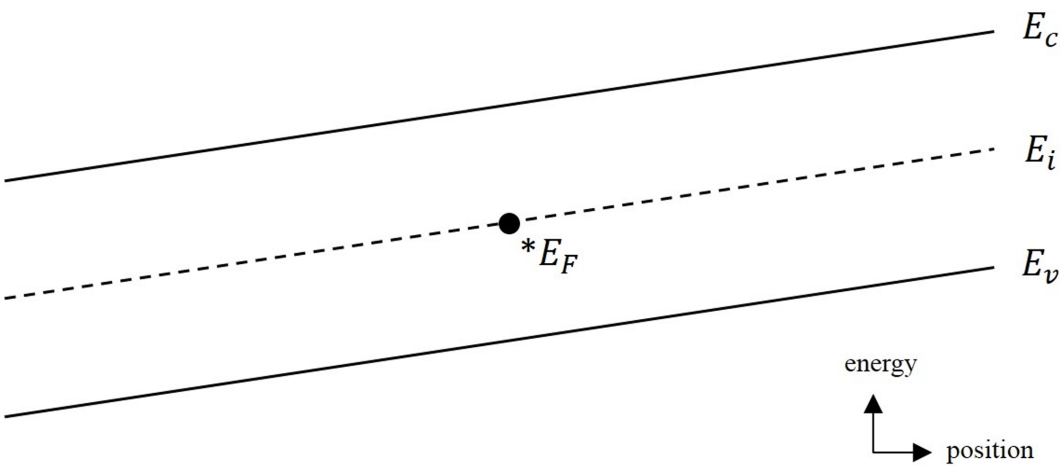


Figure 2.1: Band diagram under electric field bias

(C). Determine the direction of the electric field $\boldsymbol{E}$. Indicate the direction of **electron flows** and the corresponding **drift current** and **diffusion current** directions caused by electrons.

(D). Repeat (C) for **holes**.

3. Electron Diffusion in InP

At 300 K, a piece of InP has a majority electron concentration profile

$$n(x) = N_D e^{-x/\alpha}$$

where $\alpha > 0$.

(A). With a cross-sectional Area A_0 in the y-z plane, derive an expression for the electron diffusion current flowing through the piece.

(B). What is the electron's diffusion direction? Explain.

(C). Assume the sample is at equilibrium; calculate an expression for the electric field.

(D). The electron mobility and lifetime of InP at 300 K are 4000 $\text{cm}^2/(V \cdot s)$ and $\tau_n = 8\ \mu$s, respectively. Determine the diffusion constant D_n and the diffusion length L_n.

(E). The diffusion equation for electrons is

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\partial n}{\tau_n},$$

and the diffusion current density is

$$J_n = qD_n \frac{\partial \delta n}{\partial x}.$$

Give the definition of *steady state* and apply it to the diffusion equation to reach a simplified version which includes diffusion length.

(F). Write the general form for the solution of the partial differential equation derived in (E) as a sum of two exponentials. *Hint: see Sec. 4.4.4 in the book.*

(G). Apply the following boundary conditions to the solution derived in (F):

$$\delta n(0) = \Delta n, \qquad \delta n(L) = 0,$$

where L is the sample size. You do not need to solve these two equations; applying the boundary conditions will be enough.

4. Carrier Diffusion in p-type Silicon

In a very long p-type Si sample extending along the x-axis with cross-sectional area $A = 1\ \text{cm}^2$, electrons are injected such that the steady-state excess electron concentration is $1 \times 10^{18}\ \text{cm}^{-3}$ at $x = 0$.

(A). Given $\mu_n = 1000\ \text{cm}^2/(V \cdot s)$ and $\tau_n = 100$ ps, calculate the diffusion coefficient D_n and the diffusion length L_n of this sample.

(B). Write an expression for the excess electron concentration Δn with respect to position x.

(C). Using integration, calculate the total excess electron charge (storage charge) Q_n. Then, provide an intuitive interpretation of diffusion length.

(D). Write an expression for the electron diffusion current I_n as a function of position x. What is the maximum electron diffusion current and at what position does it occur? What is the electron diffusion current at $x = \infty$?

(E). From common sense, current in a metal wire is the same at any position (current continuity). Does your answer to part (D) violate current continuity? Why or why not? *Hint: think about the meaning of* τ_n.